P2 Chapter 11: Integration

Using Trigonometric Identities

SKILL #3: Integrating using Trig Identities

Some expressions, such as $\sin^2 x$ and $\sin x \cos x$ can't be integrated directly, but we can use one of our trig identities to replace it with an expression we can easily integrate.

Q Find $\int \sin^2 x \ dx$

?

Q Find $\int \sin 3x \cos 3x \ dx$

?

Q Find $\int \cos^2 x \ dx$

?

Q Find $\int \tan^2 x \ dx$

?

SKILL #3: Integrating using Trig Identities

Some expressions, such as $\sin^2 x$ and $\sin x \cos x$ can't be integrated directly, but we can use one of our trig identities to replace it with an expression we can easily integrate.

Q Find $\int \sin^2 x \ dx$

Do you know a trig identity involving $\sin^2 x$?

$$\cos 2x = 1 - 2\sin^2 x$$
$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$$
$$\int \sin^2 x \ dx = \frac{1}{2}x - \frac{1}{4}\sin 2x$$

Q Find $\int \sin 3x \cos 3x \ dx$

$$\sin 6x \equiv 2\sin 3x \cos 3x$$
$$\sin 3x \cos 3x = \frac{1}{2}\sin 6x$$
$$\int \sin 3x \cos 3x \ dx = -\frac{1}{12}\cos 6x + C$$

Q Find $\int \cos^2 x \ dx$

$$\cos 2x = 2\cos^2 x - 1$$
$$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$$
$$\int \cos^2 x \ dx = \frac{1}{2}x + \frac{1}{4}\sin 2x$$

Q Find $\int \tan^2 x \ dx$

Recall that $1 + \tan^2 x \equiv \sec^2 x$

$$\int \tan^2 x \ dx = \int \sec^2 x - 1 \ dx$$
$$= \tan x - x + C$$

Check Your Understanding

Q Find $\int (\sec x + \tan x)^2 \, dx$

?

Q Find $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 3x \; dx$

?

Check Your Understanding

Q Find $\int (\sec x + \tan x)^2 \, dx$

$$\int (\sec x + \tan x)^2 \, dx$$
$$= \int \sec^2 x + 2\sec x \tan x + \tan^2 x \;\; dx$$
$$= \tan x + 2\sec x + \tan x - x + C$$
$$= 2\tan x - 2\sec x - x + C$$

Q Find $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 3x \;\; dx$

$$\cos 6x = 1 - 2\sin^2 3x$$
$$\therefore \sin^2 3x = \frac{1}{2} - \frac{1}{2}\cos 6x$$
$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sin^2 3x \;\; dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{2} - \frac{1}{2} cos6x \;\; dx$$
$$= \left[\frac{1}{2}x - \frac{1}{12} sin6x\right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$
$$= \left(\frac{\pi}{6} - \frac{1}{12}\sin 2\pi\right) - \left(\frac{\pi}{12} - \sin \pi\right)$$
$$= \frac{\pi}{12}$$

Exercise 11.3

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Homework Exercise

1 Integrate the following:

a $\cot^2 x$ **b** $\cos^2 x$

c $\sin 2x \cos 2x$ **d** $(1 + \sin x)^2$

e $\tan^2 3x$ **f** $(\cot x - \operatorname{cosec} x)^2$

g $(\sin x + \cos x)^2$ **h** $\sin^2 x \cos^2 x$ **i** $\dfrac{1}{\sin^2 x \cos^2 x}$ **j** $(\cos 2x - 1)^2$

Hint For part **a**, use $1 + \cot^2 x \equiv \operatorname{cosec}^2 x$. For part **c**, use $\sin 2A \equiv 2 \sin A \cos A$, making a suitable substitution for A.

2 Find the following integrals.

a $\displaystyle\int \frac{1 - \sin x}{\cos^2 x}\,dx$ **b** $\displaystyle\int \frac{1 + \cos x}{\sin^2 x}\,dx$ **c** $\displaystyle\int \frac{\cos 2x}{\cos^2 x}\,dx$

d $\displaystyle\int \frac{\cos^2 x}{\sin^2 x}\,dx$ **e** $\displaystyle\int \frac{(1 + \cos x)^2}{\sin^2 x}\,dx$ **f** $\displaystyle\int (\cot x - \tan x)^2\,dx$

g $\displaystyle\int (\cos x - \sin x)^2\,dx$ **h** $\displaystyle\int (\cos x - \sec x)^2\,dx$ **i** $\displaystyle\int \frac{\cos 2x}{1 - \cos^2 2x}\,dx$

3 Show that $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 x\,dx = \frac{2 + \pi}{8}$ **(4 marks)**

4 Find the exact value of each of the following:

a $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{\sin^2 x \cos^2 x}\,dx$ **b** $\displaystyle\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\sin x - \operatorname{cosec} x)^2\,dx$ **c** $\displaystyle\int_0^{\frac{\pi}{4}} \frac{(1 + \sin x)^2}{\cos^2 x}\,dx$ **d** $\displaystyle\int_{\frac{3\pi}{8}}^{\frac{\pi}{2}} \frac{\sin 2x}{1 - \sin^2 2x}\,dx$

5 **a** By expanding $\sin(3x + 2x)$ and $\sin(3x - 2x)$ using the double-angle formulae, or otherwise, show that $\sin 5x + \sin x \equiv 2 \sin 3x \cos 2x$. **(4 marks)**

b Hence find $\displaystyle\int \sin 3x \cos 2x\,dx$ **(3 marks)**

6 $f(x) = 5 \sin^2 x + 7 \cos^2 x$

a Show that $f(x) = \cos 2x + 6$. **(3 marks)**

b Hence, find the exact value of $\displaystyle\int_0^{\frac{\pi}{4}} f(x)\,dx$. **(4 marks)**

7 **a** Show that $\cos^4 x \equiv \frac{1}{8}\cos 4x + \frac{1}{2}\cos 2x + \frac{3}{8}$ **(4 marks)**

b Hence find $\displaystyle\int \cos^4 x\,dx$. **(4 marks)**

Homework Answers

1 **a** $-\cot x - x + c$ **b** $\frac{1}{2}x + \frac{1}{4}\sin 2x + c$

c $-\frac{1}{8}\cos 4x + c$ **d** $\frac{3}{2}x - 2\cos x - \frac{1}{4}\sin 2x + c$

e $\frac{1}{3}\tan 3x - x + c$ **f** $-2\cot x - x + 2\operatorname{cosec} x + c$

g $x - \frac{1}{2}\cos 2x + c$ **h** $\frac{1}{8}x - \frac{1}{32}\sin 4x + c$

i $-2\cot 2x + c$ **j** $\frac{3}{2}x + \frac{1}{8}\sin 4x - \sin 2x + c$

2 **a** $\tan x - \sec x + c$ **b** $-\cot x - \operatorname{cosec} x + c$

c $2x - \tan x + c$ **d** $-\cot x - x + c$

e $-2\cot x - x - 2\operatorname{cosec} x + c$

f $-\cot x - 4x + \tan x + c$ **g** $x + \frac{1}{2}\cos 2x + c$

h $-\frac{3}{2}x + \frac{1}{4}\sin 2x + \tan x + c$ **i** $-\frac{1}{2}\operatorname{cosec} 2x + c$

3 $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 x \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) dx$

$= \left[\frac{1}{2}x - \frac{1}{4}\sin 2x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\pi}{8} + \frac{1}{4} = \frac{2+\pi}{8}$

4 **a** $\frac{4\sqrt{3}}{3}$ **b** $\frac{9\sqrt{3} - 10 - \pi}{8}$

c $2\sqrt{2} - \frac{\pi}{4}$ **d** $\frac{\sqrt{2} - 1}{2}$

5 **a** $\sin(3x + 2x) = \sin 3x \cos 2x + \cos 3x \sin 2x$

$\sin(3x - 2x) = \sin 3x \cos 2x - \cos 3x \sin 2x$

Adding gives $\sin 5x + \sin x = 2\sin 3x \cos 2x$

b So $\int \sin 3x \cos 2x \, dx = \int \frac{1}{2}(\sin 5x + \sin x) \, dx$

$= \frac{1}{2}\left(-\frac{1}{5}\cos 5x - \cos x\right) + c = -\frac{1}{10}\cos 5x - \frac{1}{2}\cos x + c$

6 **a** $5\sin^2 x + 7\cos^2 x = 5 + 2\cos^2 x = 6 + (2\cos^2 x - 1)$

$= \cos 2x + 6$

b $\frac{1}{2}(1 + 3\pi)$

7 **a** $\cos^4 x = (\cos^2 x)^2 = \left(\frac{1 + \cos 2x}{2}\right)^2 = \frac{1}{4} + \frac{1}{2}\cos 2x$

$+ \frac{1}{4}\cos^2 2x = \frac{1}{4} + \frac{1}{2}\cos 2x + \frac{1}{4}\left(\frac{1 + \cos 4x}{2}\right)$

$= \frac{3}{8} + \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$

b $\frac{1}{32}\sin 4x + \frac{1}{4}\sin 2x + \frac{3}{8}x + c$